

Chalaka Fernando



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Github



Colombo, LK

Physics graduate from the University of Ruhuna, Sri Lanka, and 2024 CERN Summer Student with research experience in particle physics, astrophysics, and machine learning. Skilled in applying computational methods to high-energy physics datasets for event reconstruction and mass prediction. Currently collaborating with researchers at CERN, the University of Ruhuna, and the Earth-Life Science Institute, while serving as a Temporary Assistant Lecturer in Physics. Interested in pursuing advanced studies in physics and contributing to international research collaborations.

Education

Bachelor of Science (Hons.) Special Degree in Physics (2019-2024)

Subject of Specialization GPA-3.75/4.00

Special Degree Course Units GPA- 3.70/4.00

University of Ruhuna

Computational Physics | Atomic and Nuclear Physics | Quantum Mechanics | Statistics | Linear Algebra

Certificate Courses

Data-Driven Astronomy Course- Coursera (2021)

Machine Learning Intermediate -Sorbonne University (2024)

Python Programming for Quantum Computing -Packt (2026)

Practical Quantum Computing with IBM Qiskit for Beginners -Packt (2026)

Advanced End-to-end QUAntum computing TEchnical Course- QURECA (2026)

Research Experience

Study of Higgs Boson mass reconstruction

(CERN Summer Project)

2024 June- 2024 August

- A dedicated deep neural network is implemented to predict the Higgs boson mass using information from assigned light leptons and other event features.

Root | Deep Learning | Python | Sci-kit | Matplotlib | TensorFlow | Keras | Jupyter

Volunteer Researcher (Remote) - Supervisor: Dr. Andre Sopczak

August 2024 – Present (voluntary)

- Collaborating remotely with CERN-based team to refine analysis methods.
- Involved in preparing follow-up results for potential future publication or presentation.

Measurements and analysis of double stars (Undergraduate research)

2023 September-2024 April

- Conducted observational research utilizing the Las Cumbres Observatory telescope on double star systems using advanced astronomical techniques under the supervision of Dr. K.V.S. Prasad and Assistant Professor Thilina Heenatigala.
- Analyzed positional and brightness data to study their physical properties and orbital dynamics.
- First author - Submitted a research paper on double star observations to the Journal of Double Star Observations (JDSO).

Astronomy | Data Analysis | Cosmology | Big Data

Weather Forecast - Github (Department of Meteorology, Sri Lanka)

- Implemented a neural network using a Recurrent Neural Network (RNN) during one month internship at the Department of Meteorology.

Machine Learning | Deep Learning | Python | Sci-kit | TensorFlow | Keras | Jupyter

Publications/ Technical reports

**Fernando, C. (2025). Astrometric Measurements of STF 1434 Double Star System
Vol. 22 No. 1 (2026): Journal of Double Star Observations**

**Fernando, C. (2024). Study of Higgs Boson Mass Reconstruction in the
Production with Two Light Leptons and a Hadronically Decaying Tau Lepton.
CERN.**

Science Outreach and Mentoring

Temporary Assistant Lecturer, Department of Physics, University of Ruhuna
(2025 June- Present)

- Conducted undergraduate laboratory sessions and lectures.
- Assisted in research projects and curriculum development.

Temporary Demonstrator, Department of Physics, University of Ruhuna
(2024 August - 2025 June)

- Conducted physics laboratory sessions and tutorial classes for undergraduate students while assisting their research projects.

Volunteer Mentor for High School Astronomy Research
(2025 – Present)

- Mentoring gifted high school students in Sri Lanka on double star observation projects (2025–Present), aiming for publication in JDSO

Creator, Univa Youtube channel.

- Developed and produced educational videos during the Covid 19 pandemic on university-level physics and mathematics in Sinhala, aimed at improving accessibility for Sri Lankan students.
- Topics include mechanics, electromagnetism, calculus, and problem-solving techniques

Extracurricular Activities

Vice President- Ruhuna Science Research Circle
(2021-2023)

- Organized the 1st International Undergraduate Research Symposium, bringing together students and researchers from diverse academic backgrounds to present and discuss their research findings.

Member of Gavel's club

(2020)

- Played a pivotal role in organizing and facilitating online English sessions for university students during the pandemic.

Skills

Programming skills: Python, Java, ROOT, latex, SQL, C++

Technical skills: Machine Learning, Deep Learning, Data Augmentation, Tensorflow, Quantum machine learning, Qiskit, PennyLane

Soft skills: Scientific writing and communication, Team collaboration

Workshops: Participated in a Quantum machine learning workshop organized by QPoland.

Reference

Dr. Andre Sopczak